NMC 2012 A-LEVEL PAPER 2 QUESTIONS

SECTION A : 5 marks each

Qn1 Suppose a and b are positive integers such that $(a + 2b)(a - b) = 10$. Find the possible value of $2a - b$.

Qn2 Without using a calculator, evaluate $\frac{1+2011^4+2012^4}{1+2011^2+2012^2}$

Qn3 A die with numbers 1, 2, 3, 4, 6, 8 on its six faces is rolled. After the roll, if an odd number appears on the top face, all odd numbers are doubled. If an even number appears on the top face, all even numbers are halved. What is the probability that a 2 will appear on the second roll?

Qn4 Let $p(x)$ be a polynomial of degree four such that $p(2) = p(-2) = p(-3) = -1$ and $p(1) = p(-1) = 1$. What is the value of $p(0)$?

Qn5 If 1, x, y is a geometric sequence and x, y, 3 is an arithmetic sequence. Compute the highest value of $x + y$.

Qn6 A car uses 8.4litres of diesel for every 100km it is driven. A mechanic is able to modify the car's engine at a cost of \$400 so that it will only use 6.3litres of diesel per 100km. The owner determines the minimum distance that they would have to drive to recover the cost of the modifications. If diesel costs \$0.80 per litre, find this minimum distance, in km.

Qn7 In a triange ABC, we have that $3\sin A + 4\cos B = 6$ and $4\sin B + 3\cos A = 1$ What is the value of angle C?

Qn8 A circle of radius 4 units is inscribed in an equilateral triangle. Given that the area inside the triangle and outside the circle is $\sqrt{3}x - \pi y$, find the value of $x + y$.

Qn9 If the roots of $x^2 - bx + c = 0$ are $\sin(\frac{\pi}{7})$ and $\cos(\frac{\pi}{7})$. What is the value of b^2?

Qn10 A written test has 30 multiple choice questions. A student will get 5 marks for each correct answer, 1 mark for each un answered question and 0 points for each wrong answer. Suppose that at the of the test, 4 students make the following statements:

Chris says "My test score is 147", Blair says "My test score is 143", Drew says "My test score is 141" and Alex says "My test score is 139"

Only one of the students could possibly be correct. Who is this?

SECTION B

Qn11 There are 6 litres of pure alcohol in container A and 6 litres of pure water in container B. An empty bottle is filled with alcohol from container A and then emptied into container B. After shaking, the bottled is filled with the mixture from container B and emptied into container A. The ratio of alcohol to water in container A is now 4 : 1. Assuming that there were no spills, what is the size of the bottle, in litres?

Qn12 A system of 5 equations in the unknowns x, y, z, u, v has a unique solution which is a permutation of numbers 1, 2, 3, 4, 5. Five participants in a Mathematics contest were asked to solve the system, but they were only able to provide the following partial answers before the end of the alloted time:

Jane: $z = 1, u = 2$; Bob: $y = 3, v = 2$; Cathy: $z = 5, u = 3$; David: $x = 4, v = 2$; Eva: $x = 4, y = 1$.

From these partial answers each of the participants had found the correct value of only one unknown. What is the value of $x + z + v$?

Qn13 Using the iterated formula that $a^{b^c} = a^{(b^c)}$; and given that x is a real number which satisfies the equation $2^{2^x} + 4^{2^x} = 42$, find the value of $\sqrt{2^{2^{2^x}}}$.

Qn14 Determine all integers x and y that satisfy $(x + 2)^4 - x^4 = y^3$.

Qn15 What is the value of x in the plane figure below?

NMC 2012 A-LEVEL PAPER 2 SOLUTIONS

SECTION A: 5 MARKS EACH

Qn1. We are given that $(a+2b)(a-b)=10$. This implies that $(a+2b)$ and $(a-b)$ are factors of 10. We know that factors of 10 are $1, 2, 5$ and 10. Let us investigate these cases.

- Case I. When $a+2b=5$ then $a-b=2$ and solving the equations simultaneously gives $b=1$ and $a=3$. Thus $2a-b=2\times 3-1=5$.
- Case II. When $a+2b=10$,then $a-b=1$ and solving simultaneously gives $a=4$ and $b=3$. Thus $2a-b=5$.

Qn2. Let $x=2011$. Then,

$$\begin{aligned}
\frac{1+2011^4+2012^4}{1+2011^2+2012^2} &= \frac{1+x^4+(x+1)^4}{1+x^2+(x+1)^2} \\
&= \frac{1+x^4+(x+1)^2(x+1)^2}{1+x^2+x^2+2x+1} \\
&= \frac{1+x^4+x^4+4x^3+6x^2+4x+1}{2(x^2+x+1)} \\
&= \frac{x^4+2x^3+3x^2+2x+1}{x^2+x+1} \\
&= \frac{(x^2+x+1)^2}{x^2+x+1} \\
&= x^2+x+1 \\
&= 2011^2+2011+1 \\
&= 4046133.
\end{aligned}$$

Qn3. The sample space is $\{1,2,3,4,6,8\}$. The probability of an odd outcome is $\frac{2}{6}=\frac{1}{3}$ and the probability of an even outcome is $\frac{4}{6}=\frac{2}{3}$.

Case I. If an odd outcome appears first, the 'new' sample space is $\{2,2,6,4,6,8\}$. Therefore p (getting a; 2; on this roll) $=\frac{2}{6}=\frac{1}{3}$. Thus prob of getting a 2 on the second roll is $\frac{1}{3}\times\frac{1}{3}=\frac{1}{9}$.

Case II. If the first roll is even, then, the 'new' sample space is $1,1,3,2,3,8$ and the prob of getting a 2 on the second die is $\frac{1}{6}$. The probability of a 2 on the roll is $\frac{2}{3}\times\frac{1}{6}=\frac{1}{9}$.

Therefore the probability of a 2 to appear on the top face is the sum of the two answers in Case I and Case II which is $\frac{1}{9}+\frac{1}{9}=\frac{2}{9}$.

Qn4. Since $p(2)=p(-2)=p(-3)=-1, \Longrightarrow 2,-2$ and -3 are roots of $p(x)=-1$. When $Q(x)=ax+b \Longrightarrow p(x)=(x-1)(x+2)(x+3)(ax+b)-1$. Using $p(1)=p(-1)=1$, we get $2=-12(a+b) \Longrightarrow -6a-6b=1$ ie putting $x=1$, and $2=6a-6b$ when $x=-1$. On solving simultaneously we get $a=\frac{1}{12}$ and $b=\frac{-1}{4}$. The polynomial is thus $p(x)=(x-1)(x+2)(x+3)(\frac{1}{12}x-\frac{1}{4})-1$ and putting $x=0$ gives $p(0)=2$.
Alt,
let $P(x)=ax^4+bx^3+cx^2+dx+e, p(2)=16a+8b+4c+2d+e=-1, p(-2)=16a-8b+4c-2d+e=-1, p(-3)=81a-27b+9c-3d+e=-1$,this means $p(1)=a+b+c+d+e=1$ and $p(-1)=a-b+c-d+e=1. p(2)-p(-2)=16b+4d=0$ and $p(1)-p(-1)=2b+2d=0 \Longrightarrow b=-d=0$. also,$16a+4c+e=-1, 16a+4c+e=-1, 81a+9c+e=-1$ so that on solving we get $e=2$. Therefore $p(0)=0+0+0+0+0+2=2$.

Qn5. $1, x, y$ is a Geometric Progression. This implies that

$$\frac{x}{1}=\frac{y}{x} \Longrightarrow y=x^2. \tag{9}$$

$x, y, 3$ is an Arithmetic Progression. This implies that

$$y-x=3-y \Longrightarrow 2y-3=x. \tag{10}$$

Solving equations (9) and (10) simultaneously gives $y=\frac{9}{4}$ or $y=1$ implying $x=\pm 1$ and $x=\pm\frac{3}{2}$. Therefore, the maximum value of $x+y=\frac{3}{2}+\frac{9}{4}=\frac{15}{4}$.

Qn6. The cost of Car's engine modification is equivalent to the cost of $\frac{400}{0.8}=500l$ of diesel, so the car would have to be driven a distance that would save $500l$ of diesel in order to make up the cost of the modification. Originally, the car consumes $8.4l$ per $100km$, and after it consumes $6.3l$ of diesel per $100km$, saving $8.4-6.3=2.1l$. Thus,to save $500l$, the car must have been driven $\frac{500}{2.1}\times 100=23809.52km$.

Qn7. $3\sin A+4\sin B=6$. Squaring gives

$$9\sin^2 A+24\sin A\sin B+16\cos^2 B \;=\; 36. \tag{11}$$

$4\sin B+3\cos A=1$. Squaring gives

$$16\sin^2 B+24\sin B\cos A+9\cos^2 A=1. \tag{12}$$

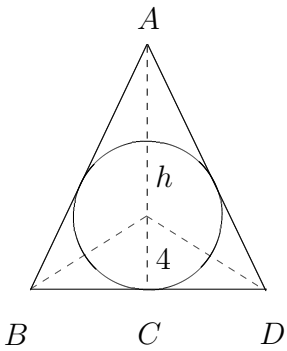
Solving equations (11) and (12) gives

$$9\sin^2 A+9\cos^2 A+24(\sin A\cos B+\sin B\cos A)+16\sin^2 B+16\cos^2 B=37.$$

$$\Longrightarrow 9=16+24\sin(A+B)=37 \Longrightarrow \sin(A+B)=\frac{1}{2} \Longrightarrow A+B=30^\circ, 150^\circ.$$

For $A+B=30^\circ$ it implies one of the angles must be less than 30°.By inspection $3\sin A+4\sin B<6$, so discard. Thus $A+B=150^\circ$ thus $<C=180^\circ-150^\circ=30^\circ$.

Qn8.



Let the height of $\triangle ABD$ be h. This implies that $\tan 30^\circ = \frac{4}{CD} \implies CD = 4\sqrt{3}$. Therefore,

$h = 4\sqrt{3}\tan 60 = 12$.

The area of triangle ABD is $\frac{1}{2} \times 12 \times 8\sqrt{3} = 48\sqrt{3}$ sq units.

Area A inside the triangle but outside the circle is $A = 48\sqrt{3} - 16\pi = \sqrt{3}x - \pi y. \implies x = 48$ and $y = 16$. Hence $x + y = 48 + 16 = 64$.

Qn9. Let the roots be α and $\beta \implies \alpha = \sin(\frac{\pi}{7})$ and $\beta = \cos(\frac{\pi}{7})$. Note that $\alpha^2 + \beta^2 = 1$, and using the given equation $x^2 - bx + c = 0$, we get $x^2 - bx + c = x^2 - (\alpha + \beta)x + \alpha\beta$ and we deduce that $b^2 = (\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta = 1 + 2c$.

Qn10. Drew is correct.

SECTION B: 10 MARKS EACH

Qn11. Let x be the size of the bottle in litres. Initially, $8l$ of pure alcohol from A are emptied into B and stirred implying that B contains a mixture of alcohol to total litres of mixture $= \frac{x}{6+x}$. xl of this mixture is emptied into A so that once again A has $6l$ of liquid. The amount of alcohol in mixture of alcohol and water in that bottle that was poured into A is $x \times \frac{x}{6+x} = \frac{x^2}{6+x}l$. This implies that A ends up with $6 - x + \frac{x^2}{6+x}l$ of alcohol and $x - \frac{x^2}{6+x}l$ of water. But the ratio of alcohol to water in A is $4:1 \implies \frac{6-x+\frac{x^2}{6+x}}{x-\frac{x^2}{6+x}} = \frac{4}{1} \implies \frac{36-x^2+x^2}{6x+x^2-x^2} = \frac{4}{1} \implies x = \frac{3}{2}$. Therefore the size of the bottle is $1.5l$.

Qn12. Assume $v = 2$, using that each contestant had found the correct value of exactly one unknown; then considering Bob and David $\implies y \neq 3$ and $x \neq 4$ and therefore considering Eva $\implies y = 1$. Since $y = 1$, then Jane's answer that $Z = 1$ is not correct, therefore $z \neq 1$ and $v \neq 2$. Using Bob and David's answers, $y = 3$ and $x = 4$. Therefore Cathy's answer $z = 5$ is correct. From

the above, since $v \neq 2$, then $v = 1$, and $u = 2 \implies v = 1, u = 2, y = 3, x = 4$ and $z = 5$. So, $x + z + v = 4 + 5 + 1 = 10$.

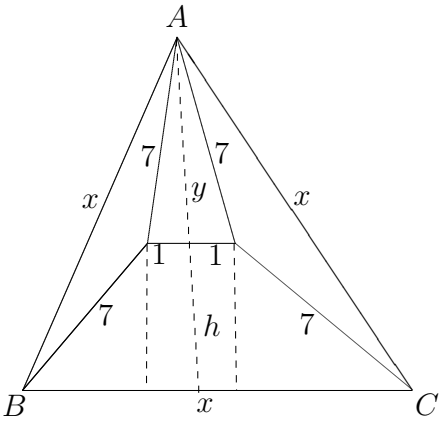
Qn13. let $y = 2^{2^x} \implies y^2 = y.y = 2^{2^x} \times 2^{2^x} = 4^{2^x} \implies 2^{2^x} + 4^{2^x} = 42 \implies y + y^2 = 42$ which on solving gives $y = -7$ or $y = 6$. Since $y > 0$, we discard $y = -7$. Therefore $y = 6.\sqrt{2^{2^x}} = \sqrt{2^y} = \sqrt{2^6} = 2^3 = 8$.

Qn14. Using the difference of two squares,

$$\begin{aligned} & ((x+2)^2 - x^2)(x+2)^2 + x^2) = y^3 \\ \implies \quad & ((x+2-x)(x+2+x))(x^2+4x+4+x^2) = y^3 \\ \implies \quad & 8(x+1)((x+1)^2+1) = y^3. \end{aligned}$$

let $a = x+1 \implies 8a(a^2+1) = y^3 \implies a(a^2+1) = (\frac{y}{2})^3$. Let $b = \frac{y}{2} \implies a(a^2+1) = b^3$. Since the R.H.S. is a perfect cube, so is a and a^2+1 for some integers q and $r \implies a = q^3$ and $a^2+1 = r^3$ and substituting for $a = q^3$ in a^2+1, we get $(q^3)^2 + 1 = r^3 \implies (q^2)^3 + 1 = r^3 + 0 \implies q^2 = 0$ and $r^3 = 0 \implies q = 0$ and $r = 1$. Substituting for $q = 0$ and $r = 1 \implies a = 0$ and $x = -1 \implies b = 0 \implies y = 0$.

Qn15.



$y = \sqrt{7^2 - 1^2} = \sqrt{49 - 1} = \sqrt{48} = 4\sqrt{3}$.

$$h^2 + (\frac{x}{2} - 1)^2 = 7^2 \implies h^2 + (\frac{x}{2} - 1)^2 = 49 \tag{13}$$

Since ABC is an equilateral triangle,

$$\sin 60^\circ = \frac{h+y}{x} \implies h + y = \frac{\sqrt{3}}{2}x.$$

But $y = 4\sqrt{3}$. Therefore,

$$h + y = \frac{\sqrt{3}}{2}x - y = \frac{\sqrt{3}}{2}(x - 8) \tag{14}$$

Substituting for h in equation (14), gives $(\frac{\sqrt{3}}{2}(x-8))^2 + (\frac{x-2}{2})^2 = 49 \Longrightarrow 3(x-8)^2 + (x-2)^2 = 49 \times 4 = 196$. On expanding and simplifying we get $4x(x - 13) = 0 \Longrightarrow$. Either $4x = 0$, or $x - 13 = 0 \Longrightarrow x = 0$ or $x = 13$. Take $x = 13$.